

Argo Technical Workshop

September 10-13, 2024

University of Washington

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Meeting Summary

The Argo Technical Workshop was held at the University of Washington in Seattle, WA on September 10-13, 2024. The goal of the workshop was to increase technical knowledge of each type of Argo float, leading to increased float lifetime and thus enhanced sustainability of the Argo array. The week included presentations by users of each type of float, the float and sensor vendors themselves, scientific motivation for sampling regimes, and interactive hands-on sessions with the floats.

This report presents a summary of the presentations and resulting recommendations and action items made by the community.

Day 1 - Tuesday, September 10, 2024

1. **Welcome, Argo Tech Community of Practice, Workshop goals (Allison Gray, Pat McMahon, Stephen Riser)**

Allison Gray and Steve Riser highlighted the importance of the Argo program, an international initiative that deploys a global fleet of robotic floats to measure ocean temperature, salinity, and pressure. Since its inception, Argo has collected over 3 million ocean profiles, significantly improving our understanding of ocean circulation, climate variability, and long-term trends in ocean heat content.

Attendees were welcomed to the University of Washington, with events including a Wednesday reception at the Ocean Sciences Building featuring a Float Lab and Lego floats. Thursday's sessions included breakout discussions, allowing participants to dive into specific technical topics and collaborative projects.

Allison thanked all participants for contributing to the workshop and for their ongoing work in advancing oceanographic research and the Argo program.

Pat McMahon gave an overview of the Argo Technician's Community of Practice (CoP) which is a collaborative network aimed at supporting Argo technicians worldwide. The CoP serves as a knowledge-sharing platform where technicians can exchange best

practices, troubleshoot issues, and improve the deployment, maintenance, and performance of Argo floats.

Pat highlighted the importance of technicians in ensuring the success of the Argo program and emphasized the need for ongoing training and development. By continuing to build an open technical community, Argo technicians can stay updated on technological advancements, operational improvements, and standardization efforts across the program.

2. MRV Vendor Presentation

MRV Systems presented their Alto and S2A Argo floats. Both floats use SBD.

S2A is the commercially available version of the SOLO 2 which was licenced from Scripps Institution of Oceanography in 2010. The current S2A is running an older version of hardware/firmware which was developed by SIO. S2A is only available with a SBE-41 CTD.

Alto was developed by MRV Systems in 2015 using the hydraulic system from the ALAMO. MRV design and software. Alto is more flexible in terms of set up and additional sensors. MRV use the Alto as their base float for R&D development projects such as microAlto, Alto-XB and Alto-BGC

Alto can be ordered with RBR or SBE CTD's. Users can also send commands directly to the CTD over iridium. MRV Systems plan to add a humidity sensor in the next 24 months.

All floats are shipped in low power (sleep) mode. A magnet is used to transition to mission mode and run a Built in Self-Test.

Testing is done via a magnet or console activated automated BIST (Built In Self Test). BIST results are transmitted via SBD which means MRV can check the results for customers as long as they are added to the SBD distribution list.

All floats are ballasted in an open tank and antennas are pressure tested to maximum operational pressure.

End of Life:

Park times can be increased to 180 days (Alto) and 3060 days (S2A) to silence problem floats.

Planned changes over the next 24 months:

Due to improvements in hardware and part obsolescence both the ALTO and S2-A will have updated hardware and software in the next 1-2 years. MRV plan to use the current SOLO-II software and electronics in the S2-A next year which includes a humidity sensor.

3. MRV User Group Presentation (Bill Dullea, WHOI)

WHOI Argo currently works with MRV built S2As and ALTO floats for the core program. Floats are received unballasted needing additional weight added before deployment. Each float is ballasted at pressure (1500 PSI max pressure) and has weight added for the deployment region. A final check is run on the float after weight has been added before it is shipped to its deployment location. A start up check (built in self test) is run on the float prior to being loaded on the ship.

We have recently added “batch testing” of the CTDs for the SBE equipped S2As. Prior to the founding of MRV Systems WHOI built SOLO I floats in house.

WHOI Argo group also uses Navis BGC floats and MRV built S2 BGC floats.

4. TWRC Vendor Presentation

Teledyne presented their APF11 Argo float (RUDICS) APF11's can be purchased with RBR L3's or SBE41 CTD's

APF11's are ballasted for specific regions. Customers are able to specify multiple target areas when they order a float and Teledyne can ballast the floats accordingly.

APF11's are shipped in IDLE (Pressure Activation) Mode which means they can't be inadvertently deployed in a frozen state.

APF11's are ballasted in a pressure vessel. Each float has been subjected to full operating pressure during ballasting.

5. TWRC User Group Presentation (Pat McMahon, CSIRO)

CSIRO deploy 50-60 Argo floats per year (Core, Deep and BGC). TWR APF11 is used primarily in areas with heavy sea ice and in shallow areas (shelf deployments) because the bladder is enclosed in a cowl.

APF11's are delivered in IDLE mode (Pressure Activation Mode) and can't be frozen. The first command a technician should issue the float is, “m_state,” to ensure the float hasn't switched to emergency mode during transit.

CSIRO has stopped opening every float and developed terra term macros (.ttl's) to automate a number of tests they run in their lab. A lot of the tests don't require special equipment (arrival test, SBE41 cracked cell test, 12 hour inflated bladder test, Shipside test)

Day 2 - Wednesday, September 11, 2024

1. NKE Vendor Presentation

NKE presented their Arvor-I Argo float which can be ordered with RBR or SBE CTD's.

All core floats are cycled in a fresh water tube to a depth of 8m before shipping. Each antenna is tested at operational pressure. A 20m test in Infremers sea water pool is optional for core floats. All Deep and BGC floats are 100% tested in Ifremer's seawater pool.

NKE perform two Zero Frequency tests on each SBE41 to detect cracked cells and check for potential sensor pollution after floats are cycled in the freshwater tube prior to shipment. The first test is conducted during the sensor integration. The second test is performed after the pool/tube test.

Floats are self ballasting by default to operate in a range of 1008 - 1038 Kg.m-3. Deployment in fresh areas requires specific ballasting and must be indicated at the time of order.

Core floats are accredited to be shipped on passenger aircraft with lithium batteries. NKE will not build floats with alkaline batteries.

The floats have two modes. ON (ready for deployment) and OFF. The buoyancy engine will pulse when the magnet is removed which signals it is in ON mode.

Users are required to connect to the float with Bluetooth and run self-test commands from a terminal program. The floats also run a self-test when the magnet is removed and "beep" to signal they have passed.

2. NKE User Group Presentation (Chris Gordon, DFO)

Argo Canada has been deploying NKE floats since late 2018, deploying a total of 206 floats of various models and sensor packages in that time. These floats have been quite reliable, especially in the context that Canada is not a huge float lab and does not test the floats much beyond their built-in self-tests. For a typical float, testing procedure is as follows: physical inspection, logged self-test and parameter changes via bluetooth connection and terminal emulator, dry run of the float launch (i.e. remove magnet, repeat self test). When possible, the self-test is repeated dockside or onboard the ship, but not always.

Additionally, Argo Canada has performed some in-water tests with the floats. This procedure is not done for all floats, as it is a large time commitment and the floats have shown to be reliable without this test. The in-water testing has been a very useful "sandbox" for learning about a new model of float and how to work with it, as familiarity

with the instruments is a crucial part of facilitating successful deployments.

Finally, the end-of-life mode on NKE floats was discussed. NKE floats enter end of life mode by changing the parameter MC0, which defines the target number of profiles for the float's lifetime, to 0 (default is 300). In the case where the float fails its self-test due to a problem with satellite visibility, subsequent attempts to launch the float will fail as it will immediately go into end-of-life mode. This will be apparent in the technical data transmitted by the float, but may present a challenge to inexperienced users. This can be fixed by resetting the MC0 parameter, either by connecting to the float directly or sending a remote command over Iridium.

3. SBS Sensor Presentation

The focus of the Seabird's presentation was on the SBE 41CP CTD, which has evolved significantly over time. The original SBE 41 supported only basic polled sampling with no internal memory. In contrast, the SBE 41CP introduced internal storage and continuous profiling, allowing for higher resolution sampling and improved data quality. Further developments led to the SBE 41N, which was designed to support biogeochemical (BGC) sensors, and the SBE 41+, which builds on this foundation while offering expanded sensor compatibility, including pH and STS. The SBE 41 Deep variant extends capabilities to 4000m depth, using a Kistler 7K P pressure sensor.

The SBE 41CP uses a Wien bridge AC circuit for conductivity measurements, ensuring stability over long deployments. Temperature is accurate to $\pm 0.002^{\circ}\text{C}$, salinity to ± 0.01 PSU, and depth to ± 2.4 dbar. To maintain long-term sensor performance, the design incorporates anti-fouling measures (AF24173) within the flow path, reducing drift over time.

A key parameter discussed was TSWAIT, which defines the delay before temperature sampling begins after the pump starts. This delay ensures proper sensor flushing and can be adjusted to optimize performance.

Handling and maintenance best practices were also discussed. Proper storage and cleaning are critical to preventing contamination and maintaining sensor accuracy. Sensors should not be exposed to temperatures above 45°C , and horizontal storage is recommended to avoid stress on the conductivity cell. The conductivity sensor itself should be cleaned with diluted bleach, Tergitol, or white vinegar, and anti-foulant devices should be replaced regularly. Before deployment, it's essential to verify serial numbers, calibration coefficients, and sensor readings to ensure all data falls within expected ranges.

We discussed Zero Frequency tests to test the conductivity sensor without a water loop. This can be done by bypassing the float controller and sending commands directly to the CTD to read the zero hertz frequency. Results need to be manually compared to the calibration sheet.

We discussed CTD power consumption—the pump typically draws 20 mA in CP mode when operating in seawater. While no major firmware or hardware changes are currently planned, the current firmware version (7.2.5) will remain the standard, with any future changes driven primarily by parts obsolescence.

A discussion on CTD performance tracking highlighted several monitoring tools, including OceanOps, as well as in-house tools developed by SBS. Looking forward, the team is considering dynamic corrections to CP data, though it remains unclear whether such changes could be implemented while maintaining backward compatibility.

Participants were encouraged to stay engaged with SBS and reach out with any questions or technical concerns via techsupport@seabird.com.

4. RBR Sensor Presentation

The RBR presentation covered key advancements in hardware, firmware, testing procedures, and fleet monitoring.

Hardware & Firmware Updates:

RBR is introducing Generation 4 (Gen4) hardware with an updated API, improved connectors, and a more compact design while maintaining sensor capabilities. The L3.5 firmware supports multiple sampling rates and schedules to facilitate a smoother transition. The PA-10 pressure sensor has been enhanced to improve long-term accuracy and stability, particularly for deep-sea deployments.

Pressure Testing & Calibration:

RBR has developed a 7,000 dbar deep pressure testing tank that allows for precise control over pressure and temperature variations. This ensures improved calibration before deployment. Additionally, the final factory testing process now includes a short air-data collection phase and verification of temperature, pressure, and conductivity readings before shipment.

Pre-Deployment Checks & Handling:

Before deployment, the instrument undergoes a pre-deployment checklist, which includes a visual inspection for any physical damage, confirmation that temperature and pressure readings are within expected ranges, and proper storage handling to prevent damage.

OEM Support & Metadata Management:

RBR has integrated JIRA for hardware and software revision tracking. A new hardware simulator has been developed to replicate sensor conditions for testing. They are also rolling out a new unified metadata format that is both human-readable and machine-

accessible, ensuring a smoother transition while maintaining legacy formats such as Coriolis.

Monitoring & Reporting Enhancements:

RBR has expanded its quality control (QC) and monitoring processes with daily checks for test failures, float status changes, and metadata validation. A Slack-based notification system has been introduced to facilitate faster issue resolution. The fleet's overall health is continuously monitored through OWC analysis, metadata validation, and compressibility checks. Additionally, monthly reports provide updates on deployments, float performance, and fleet stability.

For further questions, the RBR Argo team can be reached at argo@rbr-global.com.

Day 3 - Thursday, September 12, 2024

1. Argo Mission Timing (Stephen Riser, UW)

Note: the following summary is copied from the AST 25 report, as the presentation was repeated for the Argo Tech audience.

This talk provided an update on time-of-day sampling used by a number of floats in the Argo fleet. Several years ago it was found that a significantly high number (> 1000) of floats were configured to sample at the same time of day on each profile. In some cases this was because the floats were scheduled to surface at noon or midnight on each profile for reasons related to biological or chemical sampling issues. In other cases this was due to the floats being configured to sample with an integral number of days between profiles. Both of these cases lead to undesirable results, such as creating a bias in the estimation of heat gain in the upper ocean. An ad hoc committee set up to examine this practice strongly recommended that float surfacing be set to a non-integral number of hours for all floats deployed in the future, and that existing floats be reset to this non-integral sampling protocol to the maximum degree possible. A non-integral sampling might consist of a sampling interval of 10.08 days, so that the float will sample at all possible hours of the day over the course of several months. Other acceptable sampling schemes are also possible. Prior to the AST25 meeting, an audit of all floats that had transmitted data in the past 6 months was done. It was found that many DACs had improved the sampling protocols on their floats. However, some of the DACs with the most total floats and the most floats with sampling problems had not yet begun to change their sampling intervals. Overall, this is a serious problem and it is hoped that considerable effort will go into remedying this issue prior to the next AST meeting.

2. Discussion on Standard Core Mission including Sampling Scheme

Steve Riser from the University of Washington delivered a presentation discussing the timing and mission parameters of Core Argo floats. His talk explored the evolution of

Argo's sampling strategies, the impact of profiling cycles on ocean temperature records, and the importance of adjusting float settings to ensure uniform data collection.

He began by reviewing the original Argo mission parameters, established in 1998, which aimed for $3^{\circ} \times 3^{\circ}$ horizontal spacing (~3300 floats globally), a 10- to 14-day profiling interval, and profiling depths of up to 1500 meters. As float technology advanced, these parameters were refined to a $300 \text{ km} \times 300 \text{ km}$ grid (~3000 floats globally), a fixed 10-day interval, and a deeper profiling limit of 2000 meters. These choices were based on data from XBT lines, satellite altimetry, WOCE hydrography, and climate models.

One of the key challenges Steve addressed was the bias introduced by non-random surfacing times, which can distort estimates of ocean temperature, upper ocean heat content, and surface heat fluxes. He presented historical data on ocean temperature changes derived from Argo, showing that global ocean warming in the upper 100 meters has been estimated at $0.1\text{-}0.17^{\circ}\text{C}$ —a value comparable to mid-latitude diurnal variability ($0.1\text{-}0.2^{\circ}\text{C}$, Gille 2012). He highlighted that floats following strict 10-day cycles can introduce systematic errors, as they consistently sample the ocean at the same times of day, missing important variations in diurnal temperature cycles and heat exchange processes.

To address this issue, the Argo Sampling Committee (2021-2022) recommended an updated approach:

1. All Argo floats should sample at approximately, but not exactly, 10-day intervals—for example, 10.08 days—to create a more even distribution of surfacing times.
2. Floats should park at 1000 meters and profile to 2000 meters whenever possible.
3. Time-of-day (TOD) sampling should generally be avoided, as it leads to non-uniform sampling biases.
4. This 10.08-day interval should be implemented on all Core and Biogeochemical (BGC) Argo floats, including those deployed in the future.
5. Users should modify manufacturer defaults before deployment to ensure compliance with this protocol.
6. Existing floats following different protocols should be adjusted when possible, using two-way communication.

Steve also presented a histogram of surfacing times from a recent one-year period, showing non-random float surfacing patterns across the nine Argo Data Assembly Centers (DACs). To illustrate the effect of modifying sampling intervals, he examined data from 491 UW GO-BGC floats in the western North Atlantic, where adjusting to a 10.08-day interval (rather than exactly 10 days) improved sampling distribution and reduced systematic errors.

3. Deployment and Recovery Methods (Avery Snyder, Ocean Tech Connection)

Avery from Ocean Tech Connection (OTC) delivered a presentation on optimizing battery configurations for Argo floats and improving deployment and recovery strategies. With OTC's expertise in oceanographic engineering, he provided insights into how battery capacity, float design, and deployment methods can enhance the efficiency and longevity of Argo programs.

He began by discussing various battery configurations, ranging from 86.8Ah to 168.4Ah, demonstrating how increasing stored energy can extend float life and improve program efficiencies. However, he emphasized that simply adding more battery capacity does not necessarily lead to more profiles—floats must reach natural battery depletion for the added energy to translate into additional data collection cycles. This insight is crucial for float manufacturers looking to maximize the performance of their instruments.

Beyond battery optimization, Avery covered deployment and recovery techniques for Argo floats. He introduced a single-loop deployment method, developed by OTC and now available for purchase. This method simplifies deployment, reducing handling risks and improving reliability in the field. For float recovery, he showcased custom recovery handles for EM floats, providing a safer and more efficient way to retrieve instruments. He also shared images of reflective tape applied to CTDs and antennas, a practical enhancement designed to aid recovery in low-light conditions.

Avery highlighted OTC's range of services, including ocean instrumentation support, custom development, and field operations. With a deep understanding of oceanographic technology and practical field applications, OTC continues to support the industry by offering innovative solutions for battery management, deployment, and recovery of Argo floats.

Day 4 - Friday, September 13, 2024

1. Vendor follow-up

Each vendor answered any remaining questions from earlier in the week including technical details on profile timing.

2. Next Generation NKE ARVOR

NKE presented the next generation of the ARVOR float which includes significant firmware/software upgrades, while the float remains structurally very similar to the previous version.

3. Failure Mode Analysis of SOLO II and deep SOLO (Natalie Zilberman, SIO)

Note: the following summary is copied from the AST 25 report, as the presentation was repeated for the Argo Tech audience.

SOLO-II floats show great performance with a small (12%) overall failure rate (sum of float and CTD) over the past 7 years. Scripps is investigating cases of flooding in collaboration with SeaBird. A new antenna design (collaborative work between IDG and Maxtena) is under testing to reduce cases of float failure. Scripps is planning to run an analysis of the rate of float failure versus age (hazard rate) in order to further increase the SOLO II performance.

Deep SOLO floats show great performance with float-related failure rates similar to SOLO II over the past 7 years. Overall (SBE61+float) failure rate over the past 7 years of the Deep SOLO is 19%, close to the SOLO II (12%). Sources of failure of the SBS61 CTD (with Keller) are well understood. Recommendation for SeaBird is to install a humidity sensor on the SBE61 and SBS61 to better understand causes of CTD failure. Scripps is planning to deploy 3 Deep SOLO floats in 2025 with improved Keller and better integration on the SBS61. IDG is working on the implementation of a glass radome on Deep SOLO antennas to increase float longevity under ice.

4. Creation of Recommendations & Action Items (Rick Rupan, meeting conveners)

Recommendations and action items were discussed as a group. See below for details.

Recommendations & Action Items

1. Maintain community communication. The Argo Tech group will meet quarterly in virtual meetings and bi-annually in person.

The Argo Tech asks the AST to promote involvement in the Argo Tech community through quarterly and bi-annual meetings to all programs deploying Argo floats, including/especially smaller programs. This will also be communicated to the data management and DMQC teams to invite those interested to participate.

The slack channel used during this meeting will be converted to a long-term workspace and Argo Tech will ask OceanOps to update/use an Argo Tech mailing list which already exists, but is out of use and will be updated with contact information from this meeting and our less formalized mailing list. Manufacturers/vendors will be included in both places, as the open communication between user groups and vendors was an important success of this workshop.

2. Data capture on pre-deployment testing: use binary or numerical scale to represent the level of pre-deployment dockside testing performed on a float.

This will encourage testing and allow for analysis of float failures in context of knowing the float was in working order as close to deployment as possible. Data capture methodology to be determined. Meta file may be a good place to store this data, however Argo Tech recognizes the challenges associated with additions to the data system.

3. Improved sharing of post-deployment logs with vendors.

Transmitted data and engineering log info should be sent to vendors so that they may monitor float performance as well. For SBD, simply add the preferred vendor email to the distribution list. For floats on RUDICS communication, collaborate with vendors to arrange appropriate access.

Argo Tech asks for support as needed from AST/ADMT as DACs may need to facilitate.

Argo Tech asks vendors to review and analyze data as it is submitted.

4. Create default profiling regimes. This includes cycle timing (not equal to integer number of days), vertical resolution, and park depth sampling.

This refers to the default setup of a float delivered by vendors to a user that has not requested any specific mission customization. For this less experienced user, it represents a profile in line with modern sampling strategies and telemetry capabilities. For larger/more experienced groups, this does not represent a set of parameters they must adhere to.

With the ASTs approval, Argo Tech will work with vendors to create config files that adhere to a “default” Argo profile with a cycle period of 10.08 days, 1 dbar bins above 10 dbar, 2 dbar bins in the rest of the water column, and P/T measurements at 1000 dbar park depth every 6 hours. For groups where this may represent a volume of data in excess of their telemetry capabilities or budget, we suggest reducing resolution below 1000 dbar as necessary (10, 20, 25 dbar bins etc.).

This addresses action item 9 from AST 25: *AST suggests adding a topic to the float technical workshop to consider providing a default Argo mission configuration file, by float type. This could include shallowest pressure bin, high resolution sampling in top 10m, a cycle time of 10.08 days, hourly drift sampling, TSWAIT for SBE CTDs etc. TcoP chairs, float experts, AST members.*

5. Enhance understanding of float failure modes.

Larger user groups should track float failures and present analysis at bi-annual Argo Tech meetings and/or updates at quarterly virtual meetings (see SIO failure analysis presentation as an example). Any systematic failures should be communicated to the vendors and the Argo Tech group to prevent deployment of compromised floats.

6. Improve fault identification and record-keeping.

This action item is a call for improved record keeping prior to float deployment. This includes logging all interactions with the float, for example using date-stamped terminal logs. In addition, any problems discovered during float testing/ballasting/assembly should be formally recorded by the end user and shared with the vendor, even in cases where the problem can be fixed without any required RMA or similar action from vendors. WHOI maintains a github for this type of information, which is a good example to follow.

7. Map appropriate float parameters to meta file variables, manufacturers and users log during checks.

Float parameters that directly map to parameters in Argo meta files should be included in float check logs as well, by the manufacturer and/or in the log created by the user during testing. Argo Tech will work with manufacturers to map the language of these parameters to Argo parameter names across different platforms.

This action is complementary to the ADMT action [#26](#) to make sensor parameters/calibration coefficients machine-readable via JSON file. See more here: [euroargodev/sensor_metadata_json_at_dev/0.4.0](https://euroargodev.github.io/sensor_metadata_json_at_dev/0.4.0)

8. Recommend best practices for float testing.

Previously published Ocean Best Practices for Argo Floats (Morris et al. 2024, <https://doi.org/10.25607/OBP-1967>) describes testing Argo floats as a best practice, but Argo Tech can provide further granularity to these tests. This could include what tests, when and how often to run them, engineering data to look at, a template for recording faults as described in items 5 and 6, etc.

Argo Tech will assemble a float testing best practices document and seek review from appropriate sources such as the AST and/or authors of the published best practices paper.

9. Energy budget determination for float missions

Argo Tech requests vendors develop enhanced tools for planning, analyzing float energy

budgets. The existing tool developed by NKE for the PROVOR CTS5 is a good example.

10. Include humidity sensors in floats for leak detection.

Not all floats are equipped with humidity sensors for leak detection. All vendors plan to release controllers with humidity sensors on next generation floats to enhance leak diagnosis and failure analysis.

11. Float depth control and pressure sensor requirements for end of life.

All manufacturers can send commands to keep a float at depth, however some floats require a functional pressure sensor to leave the surface. Argo Tech requests all floats be capable of being sent below the surface for end of life in the event of CTD sensor failure.

12. How can users efficiently access manufacturer test results?

Most tests performed by the manufacturer can be accessed by the user through telemetry, as the user group typically activates Iridium service before delivery. Other tests are sometimes stored in the memory of the float itself. Argo Tech will work with manufacturers to clarify where and how to access these tests, and request adjustments where appropriate.

13. 0 Results for a zero frequency test stored on board SBS CTD sensors in the same way as calibration coefficients

Additionally, Argo Tech requests the 0 frequency test data be stored in SeaBird onboard float information, and that the 0 frequency test can be easily re-run through connection to the float, perhaps as part of the default self test. It should be noted however that this test would fail if the CTD is wet.

14. Clarification of SBS commercial and hardware nomenclature.

SBS presented some clarification on hardware nomenclature, which can vary depending on float type. This should not be confused with the commercial name of the SBS CTD. Action for the hardware details be captured in the metadata via upcoming standardized machine readable JSON metadata to be provided with the sensor. This will require collaboration with the data team on how best to store this information.

15. Temperature tracking throughout shipping.

Due to shipping in IDLE (pressure-activation) state, TWR APEX floats periodically check pressure and temperature. The data is stored on board and accessible by the end user,

resulting in a temperature record throughout the shipping process. Argo Tech would request this from other vendors as well as it could be a useful diagnostic record. SBS noted a desired temperature during shipping $<45^{\circ}\text{C}$; there is currently no way to evaluate how many sensors may be experiencing those temperatures prior to deployment across the whole array.

16. Enhanced language surrounding float performance.

Argo Tech encourages members of the Argo community to carefully frame discussion of float failure. Keep in mind complimentary metrics like cost per profile or data when appropriate. Additionally, floats that last their expected lifetime and reach end of life though battery exhaustion should not be considered “failures”.

17. Ability to check voltage on individual battery packs without opening the float on future platforms or alternative method to verify all packs are connected to the controller

Verifying voltage on each pack ensures floats are not deployed with any bad batteries, reducing uncertainty when a float does not reach its expected number of profiles and allowing failures to be caught and rectified before deployment.

18. Argo Tech will send a representative to the UNOLS RVTEC (Research Vessel Technical Enhancement Committee) meeting annually.

Especially in the US, UNOLS technical staff deploy floats on a regular basis, often during missions not related to Argo and at no cost. Argo Tech wishes to send a representative to the yearly RVTEC meeting to (1) present on best practices for deploying Argo floats from an oceanographic vessel, (2) express the Argo program’s appreciation/gratitude for their contribution to the program and (3) work with technicians and ship operators for better coordination to ease the burden for deploying floats.

19. User groups will use ticketing systems provided by vendors for all float issues. Manufacturers will ensure their responsiveness.

User groups, especially those with close relationships with float vendors or sensor manufacturers, will often bypass the typical customer service portals these companies have in place in favor of direct contact with a representative they may have a personal relationship with. Bypassing these types of ticketing systems, however, can lead to issues not being properly tracked internally by the vendors.

User groups should be motivated to use these ticketing systems as it is the best way for vendors to track issues, and make changes to their products based on that information. It is a way for both end users and vendors to contribute to the progress/reliability of a given float/sensor.

On the manufacturer's side, reliability of these generalized email addresses will help in ensuring widespread adoption by users, as quicker, more reliable responses from personal connection discourages use of the ticketing system.

Overall, a mix of emailing manufacturers directly and copying their general support email is the suggested practice that should satisfy all parties.

20. Track the above action items in an open format using github

Model after the ADMT action item dashboard ([All ADMT actions · ADMT - actions dashboard](#)). Check with ADMT co-chairs if it would be preferable for Argo Tech to host their own organization or if it would be more appropriate to set it up under the OneArgo github organization.